A counterexample in every band of the parabola conjecture

A solution to MathDB 347624

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Abstract

Vielhaber's proposed quadratic lower bound for the inverse binary question-mark function is false. At $y = 7/8$ the inverse is $4/5$, whereas the proposed lower bound is $13/16$. More generally, the bound fails by exactly $1/(80 \cdot 2^k)$ at $y = 7/2^{k+3}$, for every integer $k \geq 0$. All evaluations follow directly from the defining finite binary codes.

Task 1: the original problem and its full solution

Write $\phi = \overline{?}_V^{-1}$ for the real-valued inverse in Definition 2(iii) of [1]. Conjecture 22 of that source asserts that, whenever $k \geq 0$ and $2^{-k-1} \leq y \leq 2^{-k}$,

$$\phi(y) \geq Q_k(y), \qquad Q_k(y) = \frac{4}{3}\,2^k \left(y - \frac{3}{4}\,2^{-k}\right)^2 + y - \frac{1}{12}\,2^{-k}. \tag{1}$$

It also specifies equality at the two band endpoints and its three-quarter point. We disprove the inequality itself at interior points distinct from all three specified points.

The coding convention

For a positive integer $b = 2^\ell + \sum_{j=0}^{\ell-1} a_j 2^j$, where $a_j \in \{0, 1\}$, the source defines

$$C_{\mathrm{I}}(b) = 0^\ell 1 \overline{a}_{\ell-1} \cdots \overline{a}_0, \qquad C_{\mathrm{II}}(b) = 1^\ell 0 a_{\ell-1} \cdots a_0, \qquad \overline{a} = 1 - a.$$

An exponent on a bit denotes repetition; an exponent zero denotes the empty word. If $r = [b_1, \ldots, b_{2m}] \in (0, 1)$ is an even-length continued fraction, in the convention $[b_1, b_2] = 1/(b_1 + 1/b_2)$, concatenate

$$w = C_{\mathrm{I}}(b_1) C_{\mathrm{II}}(b_2) \cdots C_{\mathrm{I}}(b_{2m-1}) C_{\mathrm{II}}(b_{2m}).$$

Remove all trailing zeroes and then the final one to obtain the address v. The corresponding dyadic argument is $y = (0.v1)_2$, and $\phi(y) = r$. Equivalently, $y = (0.w)_2$, since removing trailing zeroes does not change the binary fraction. This is Definition 2 and the address identification in Appendix 2 of [1].

The first counterexample

The even-length expansion $4/5 = [1, 4]$ has

$$C_{\mathrm{I}}(1) C_{\mathrm{II}}(4) = 1\,11000 = 111000.$$

Its address is $v = 11$, so $\phi(7/8) = 4/5$. Independently, Figure 4 of the source gives precisely this address, argument, and value. But

$$Q_0(7/8) = \frac{4}{3}\left(\frac{1}{8}\right)^2 + \frac{7}{8} - \frac{1}{12} = \frac{13}{16} = \frac{4}{5} + \frac{1}{80}.$$

Thus (1) is false.

An exact failure in every band

Theorem 1. *For every integer $k \geq 0$, put $y_k = 7/2^{k+3}$. Then*

$$\phi(y_k) = \frac{4}{5 \cdot 2^k}, \qquad Q_k(y_k) = \frac{13}{16 \cdot 2^k}, \qquad Q_k(y_k) - \phi(y_k) = \frac{1}{80 \cdot 2^k} > 0.$$

Proof. The case $k = 0$ was proved above. For $k = 1$,

$$\frac{2}{5} = [2, 2], \qquad C_{\mathrm{I}}(2)C_{\mathrm{II}}(2) = 011\,100, \qquad (0.011100)_2 = \frac{7}{16}.$$

Now let $k \geq 2$ and set $b = 5 \cdot 2^{k-2} - 1$. Since $2^k \leq b < 2^{k+1}$, its code uses $\ell = k$. Moreover,

$$b - 2^k = 2^{k-2} - 1.$$

Written with exactly k bits, this last integer is $00\,1^{k-2}$; its bitwise complement is $11\,0^{k-2}$. Hence

$$\frac{4}{5 \cdot 2^k} = \frac{1}{b+1} = [b, 1], \qquad C_{\mathrm{I}}(b)C_{\mathrm{II}}(1) = 0^k 1110^{k-2}\,0.$$

The associated dyadic value is $2^{-k-1} + 2^{-k-2} + 2^{-k-3} = 7/2^{k+3}$, proving the inverse evaluation. In every case the address is $0^k 11$.

Since $y_k = (7/8)2^{-k}$, it lies strictly inside the required band. Substitution gives

$$Q_k(y_k) = 2^{-k}\left(\frac{4}{3}\left(\frac{1}{8}\right)^2 + \frac{7}{8} - \frac{1}{12}\right) = \frac{13}{16 \cdot 2^k}.$$

Subtracting $4/(5 \cdot 2^k)$ gives the stated positive gap. □

This resolves the conjecture negatively. The obstruction uses its stated coding convention and occurs in every band, including $k = 4$ mentioned in the source. It does not depend on an endpoint interpretation, a numerical approximation, or the existence of the extension away from dyadic arguments.

Task 2: the least uniform vertical correction

Determine the smallest $c \geq 0$ for which $\phi(y) \geq Q_k(y) - c2^{-k}$ throughout every band. The theorem forces $c \geq 1/80$. Obtain a matching upper bound or identify larger obstructions and the points approaching the supremum.

Task 3: the limiting shape of small bands

For $1/2 \leq t \leq 1$, study $F_k(t) = 2^k \phi(2^{-k}t)$. Determine its limiting profile and a uniform error estimate, using the first partial denominator and the remaining code. The exact identity $F_k(7/8) = 4/5$ supplies a consistency check.

Task 4: the set where the quadratic bound fails

For each k, describe the set of $t \in [1/2, 1]$ satisfying $F_k(t) < \frac{4}{3}(t - 3/4)^2 + t - 1/12$. First classify dyadic arguments by their finite codes, then determine the interval structure and measure of the full failure set.

Task 5: sharp multiplicative bounds

Determine $\inf_{0<y<1} \phi(y)/y$ and $\sup_{0<y<1} \phi(y)/y$, with their equality cases. Compare the result with the separate linear-bound Conjecture 21 of [1]. The present counterexamples have the constant ratio $32/35$ and do not disprove its proposed lower ratio $8/9$.

Verification and status. The accompanying standard-library Python verifier uses exact rational arithmetic and separately implements coding and decoding. Its finite checks supplement the proof for every k. Tasks 2–5 are proposed directions and receive no solution credit. The full 20-page primary source and the documented bounded prior-claim search were checked by the authoring agent; there has been no independent peer review.

References

[1] M. Vielhaber, *V Tree—Continued Fraction Expansion, Stern–Brocot Tree, Minkowski's ?(x) Function In Binary: Exponentially Faster*, arXiv:2008.08020v1 (2020), Definitions 1–2, Conjecture 22, and Figure 4. `https://arxiv.org/abs/2008.08020`.